

Bridging the Digital Divide in Public Health: A Comparative Evaluation of Retrieval-Augmented Generation for Universal Health Coverage Intelligence (UHC) in LMIC Nations

Background: Low-middle-income countries face persistent digital health divides impeding UHC attainment under SDG 3.8. WHO Health Policy Systems Research (HPSR) identifies need for quantitative performance assessment with qualitative governance frameworks to generate actionable health intelligence.¹ While retrieval-augmented generation technologies offers transformative potential to democratize health intelligence, deployment in resource-constrained settings to address governance challenges - reliability, equity and algorithmic accountability.

Objective: To evaluate open-source RAG platforms for reliability, equity implications, governance requirements for UHC policy intelligence and develop an empirically governance framework.

Methods: A pragmatic mixed-methods embedded design evaluated two platforms (AnythingLLM's; RAG Flow), across two language models (Llama 3.2 3B; Gemma 3 27B) under two parameter configurations (permissive vs. restrictive) using WHO SDG documents (2005–2025). We assessed 200 query-responses for factual accuracy, hallucination rates, protocol adherence, and response consistency (Fleiss' $\kappa=0.82$) using WHO ground truth benchmarks to conceptualize RAG-PHI governance framework using context-coding.²

Results: Restrictive configurations reduced hallucination by 62.5% for RAG Flow with Llama 3.2 3B (Cohen's $d = 0.89$; $F[1,7992] = 384.2$, $p < 0.001$; $\eta^2 = 0.046$), achieving highest factual accuracy ($M = 2.31$, $SD = 0.67$; $ICC = 0.97$). Critical failures included Gemma 3 27B generating fabricated India's UHC Service Coverage Index (88.9 vs. WHO-verified 54.3) across both configurations, RAG Flow non-response to 85% of UHC queries under permissive settings and AnythingLLM's response latency variance of 30-fold (500 ms to 15 s) for identical queries.

Discussion: Configuration insensitivity in Gemma 3 27B reveals a training-corpus governance failure unresolvable by parameter tuning, generating cross-layer insights unavailable from either quantitative or qualitative analysis independently.

Conclusion: Current RAG implementations requires substantial improvement before responsible LMIC deployment utilizing proposed conceptualized RAG-PHI framework (ethics-as-infrastructure) operationalized empirically across three layers: knowledge-based sovereignty, configuration accountability, deployment equity. Future research should evaluate multilingual capabilities, SMS/voice modalities for deployment across diverse LMIC health systems.

Keywords: Retrieval-Augmented Generation; Universal Health Coverage; digital health equity; health information systems; artificial intelligence; low- and middle-income countries; responsible AI; SDG 3.8; public health informatics

1.0 Introduction

The global pursuit of Universal Health Coverage under Sustainable Development Goal 3.8 confronts a fundamental paradox: two decades of WHO reporting (2005–2025) have generated vast health datasets, yet the administrators who most urgently require these insights for policy formulation often cannot access them. A district health officer in rural Bihar managing over 600 public healthcare centres may spend 24–48 hours manually aggregating data across repositories before answering a fundamental question as to how many children received measles vaccines in their district over three months which implies not a data scarcity problem but a health intelligence access problem.

WHO DHSPR 'Changing Mindsets' strategy argues that health policy and systems research must function as 'the GPS of health decision-making' to enhance navigational support that locates the health problem, identifies the desired outcome, and maps feasible routes to improvement.¹ LMICs face structural barriers that impede this navigation: fragmented information architectures, HR shortages, limited digital infrastructure, and linguistic diversity that renders most AI health tools inaccessible to non-English-speaking populations.^{3–5}

Retrieval-Augmented Generation (RAG) represents a promising approach to bridging this divide by grounding language model responses in verified external documents rather than relying solely on pre-trained knowledge. A well configured RAG system retrieves relevant passages from a curated document collection before generating a response unlike standard chat bots, which may hallucinate plausible-sounding but factually incorrect information.⁶ There are very few sparse studies on the systematic evaluations of RAG platform reliability for public health intelligence in the context of LMIC nations. This study analyzes the three research objectives which are framed to inculcate the pragmatic embedded mixed-methods design quantitatively, qualitatively and integration of the mixed method approach:

RQ1: How do platform, language model, and parameter configuration affect factual accuracy, hallucination rate, protocol adherence, and response consistency in open-source RAG systems evaluated against WHO UHC ground-truth data?

RQ2: What empirically-derived governance framework based on the failure-mode patterns of RAGFLOW and Anything LLM can responsibly guide RAG deployment for public health intelligence across LMIC settings?

RQ3: How do quantitative configuration-performance findings, when systematically integrated with implementation science frameworks, generate governance insights that neither of the methodological approaches could independently generate?

The above emphasizes to congregate on a central governance proposition: RAG configuration choices are not technically neutral technical decisions but they act at the infrastructure-level governance which has direct equity consequences for the health administrators in LMIC nations.

2.0 Literature Review

2.1 Healthcare AI and RAG Systems : Recent studies on RAG in healthcare have highlighted the potential of grounding large language models (LLMs) in curated evidence bases to mitigate hallucinations in clinical and biomedical contexts.⁷⁻⁸ RAG-powered health intelligence systems must serve dual purposes in LMIC contexts: supporting high-precision policy analysis (e.g., 'How has out-of-pocket health expenditure changed across LMIC nations between 2015–2023?') and providing accessible citizen guidance on health policies, facilities, and schemes.⁹⁻¹⁰

Traditional decision support systems excel at pre-structured queries against defined data warehouses but fail to meet ad-hoc analytical demands.¹¹⁻¹² RAG addresses this limitation by grounding LLM responses in verified external information,⁶ which is critical in healthcare where hallucinated information poses direct risks to patient safety and public health decision-making.⁷

2.2 Governance Frameworks for AI in Public Health: WHO HPSR strategy establishes that health systems are highly complex, multi-layered, and nonlinear, requiring research designs that combine quantitative and qualitative methods to assess interventions.¹ HPSR strategy argues that health policy and systems research must transcend individual discipline boundaries and function as trans-disciplinary, demand-driven inquiry embedded within decision-making ecosystems.

Rogers et al.'s integrative context coding framework, integrated with Consolidated Framework for Implementation Research (CFIR), offers a systematic approach to evaluating the multi-level contextual determinants impacts the implementation outcomes across public healthcare systems.² The framework consolidates diverse evidence sources such as organizational documents, interview data, and observational records organized as a structured analytical mechanism which suggests contextual influences across varied levels such as individual, team, organizational, and system. This study applies analogous multi-level reasoning to RAG governance, positioning configuration choices, knowledge base architecture, and deployment modality as contextual determinants of implementation equity and not solely as incidental technical parameters. The existing global frameworks offer only partial coverage as explicitly stated in WHO (2021) framework and the OECD AI Principles which only establishes useful normative benchmarks but does not address RAG-specific governance in resource-constrained healthcare deployments across LMIC nations.¹³ Dynamic AI retrieval systems operationalization across varied healthcare settings of the LMIC nations are not addressed by the foundational guidance of the CARE and FAIR Data Principles.

2.3 Research Gaps: A PRISMA-guided search across varied databases such as PubMed (n=182), Embase (n=121), and PsycInfo (n=32) resulted in identifying 335 studies, of which 20 met inclusion criteria following de-duplication and screening as shared in additional supplements

<https://github.com/vinithaFPM23005/-FP05-Bridging-the-Digital-Divide-in-Public-Health.git>.

Four substantive gaps emerged: (i) a persistent clinical-versus-policy imbalance, with 89% of LLM studies concentrated on clinical applications and none evaluating RAG for SDG monitoring or UHC tracking;⁷ (ii) reliance on single-platform designs, with no cross-platform comparative evaluation of open-source systems under identical conditions;^{14–15} (iii) acknowledgment of parameter effects without systematic quantification;^{16–17} and (iv) LMIC-oriented conceptual roadmaps that lack grounding in empirical deployment evidence.¹⁸

Theoretical and Conceptual Framework: This study grounds its governance framework in two complementary theoretical foundations: the WHO HPSR strategy's argument that health systems research must embed evidence generation within decision-making ecosystems,¹ and the context coding framework stated in Rogers et al.'s establishes that implementation success depends on systematic multi-level assessment of contextual determinants.² WHO HPSR and Rogers et al. frameworks in consensus arrive at a shared core principle: governance cannot be retrofitted as downstream compliance, but it must be embedded within data architectures, configuration regimes, and institutional arrangements from inception. When applied to RAG-based health systems, this means that infrastructure choices governs similarity thresholds, temperature settings, retrieval depth, and embedding strategies collectively determine which queries receive reliable responses and whose health realities are rendered legible within the system and which population it is catered to. This study present findings to substantiate that the hallucinations observed were not random model errors but predictable consequences of design and training choices.

3.1 The RAG-PHI Conceptual Framework: The literature review provided a theoretical foundation wherein which this study conceptualizes to propose the Retrieval-Augmented Generation for Public Health Intelligence (RAG-PHI) framework as its primary conceptual contribution. RAG-PHI articulates three interlocking governance layers derived empirically from the factorial experiment's failure-mode analysis:

Layer 1: Knowledge-Base Sovereignty: The corpus is treated as a governance artefact, not a neutral repository. Document selection, metadata enrichment, chunking strategy, and quality control shape representational equity. In LMIC deployments, national and sub-national data – including district-level KPIs and indigenous health knowledge – must be prioritized alongside global WHO reports.

Layer 2: Configuration Accountability: Parameters such as temperature, similarity threshold, top-K retrieval, and penalty terms are governance levers that must be documented, justified, and verifiable. For high-stakes policy queries, restrictive configurations that favor precision and explicit 'I don't know' responses over fluency are normatively required.

Layer 3: Deployment Equity: Responsible deployment demands multilingual interfaces, infrastructure-appropriate modalities (SMS and voice for low-bandwidth settings), transparent uncertainty communication, and phased roll-out in which human-verified support precedes partially automated decision-support. RAG-PHI aligns with the WHO HPSR strategy's call for governance frameworks that are 'embedded within decision-making processes' and the Rogers et al. framework's emphasis on monitoring contextual dynamics longitudinally across system levels.¹⁻

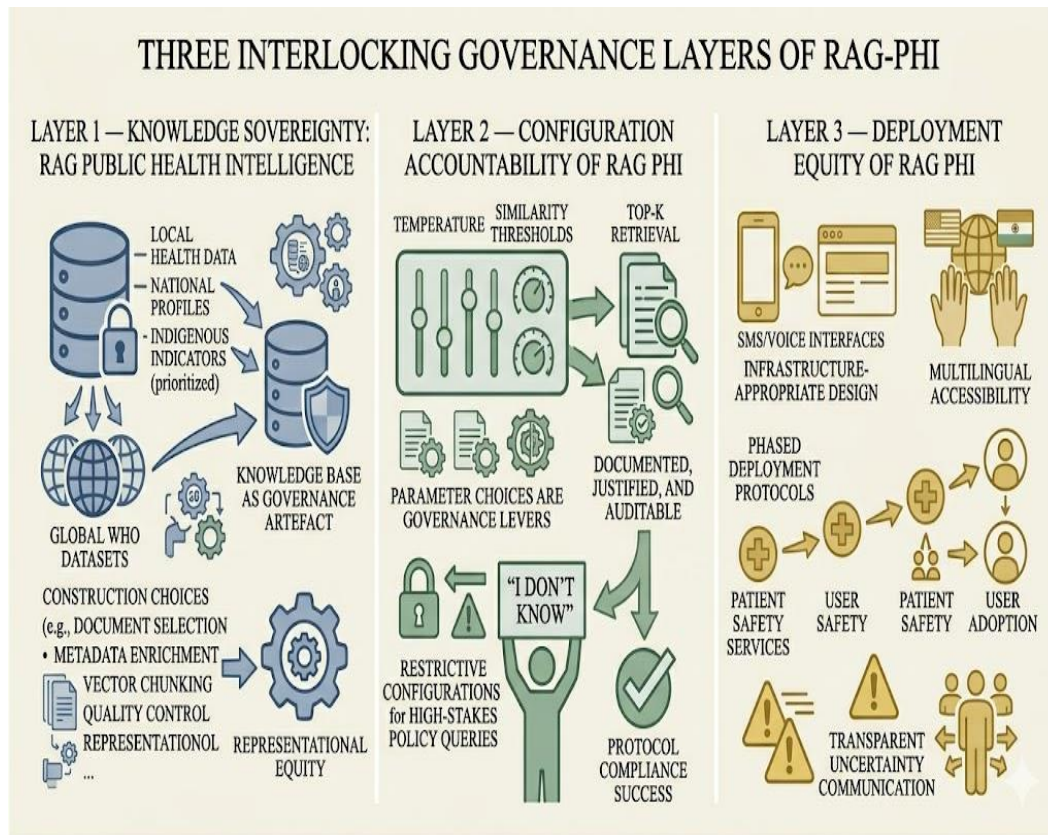


Figure 1 : The RAG-PHI Conceptual Framework

(Source: Author's own created using Gemini AI for image generation only)

3.2 Theoretical Framework Variables:

RAG-PHI governance framework integrates three theoretical domains through four independent pillars influencing system efficacy via three mediating process variables. Contextual factors moderate these pathways, optimizing four dependent outcomes while controlling for corpus size, model architecture, and prompt structure. { (Tables 1 and 2 operationalize the framework variables and is provided in Supplementary Details available at: <https://github.com/vinithaFPM23005/-FP05-Bridging-the-Digital-Divide-in-Public-Health.git> as Table_S1: Supplementary S1 and Table_S2 as Table_S2: Supplementary S2) } whereas Figure 2 shows theoretical framework for the full operationalization of the RAG-PHI framework variables and construct definitions mapped with Independent, Dependent, Moderating, Mediating, and control variables using RAG-PHI, System Configuration and Parameters of the models as shown below.

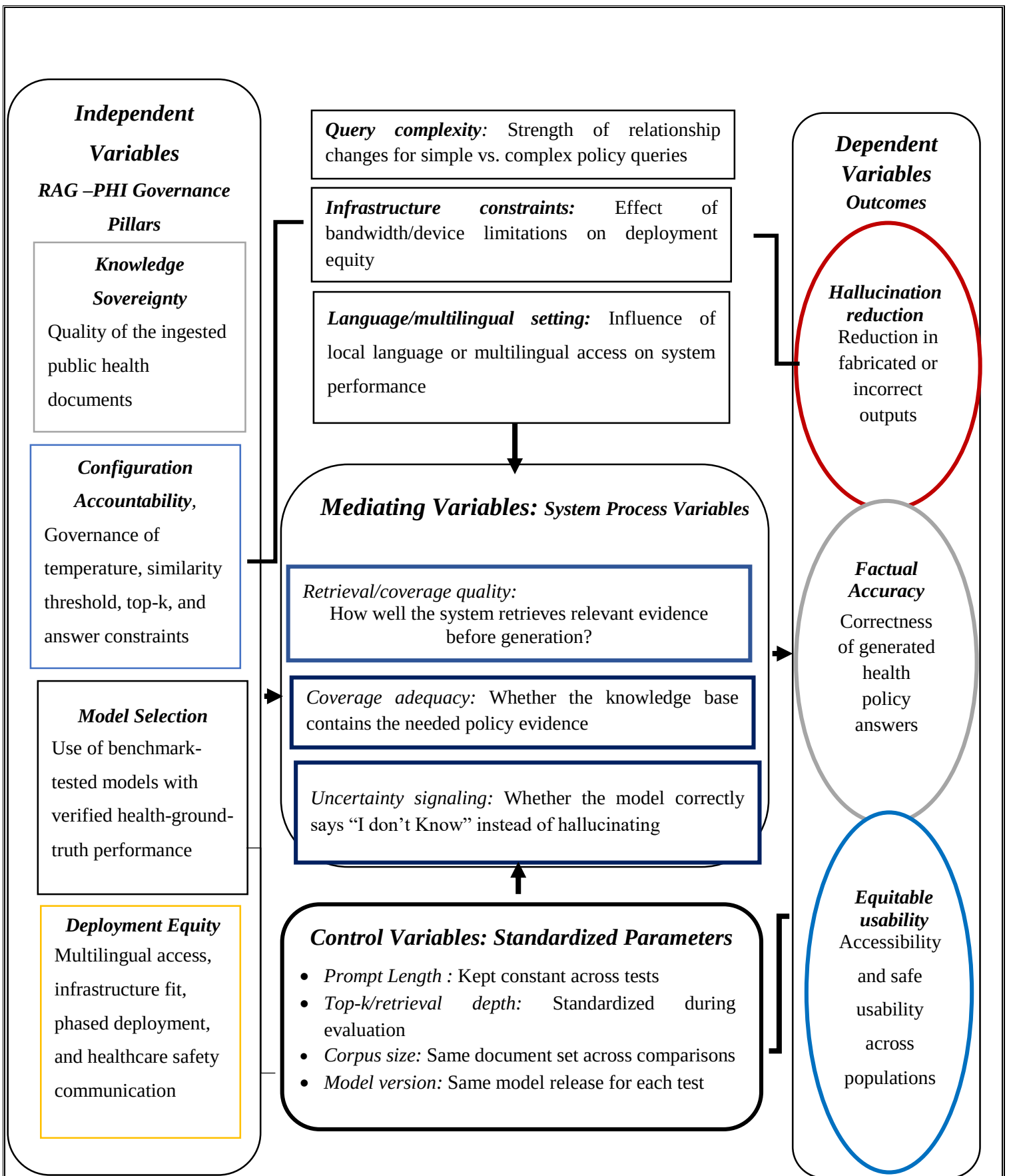


Figure 2. Proposed Conceptual Theoretical framework using RAG PHI, System Config and Parameters of RAG PHI (Source: Author's Own Work4. Methodology)

4.0 Methodology: A brief Methodology Summary as per JARS Mixed Items Checklist is detailed in Table 1

<i>Design Element</i>	<i>Quantitative Component</i>	<i>Qualitative/Conceptual Component</i>
<i>Purpose</i>	Evaluate RAG platform reliability for UHC policy intelligence	Develop RAG-PHI governance framework grounded in empirical findings
<i>Approach</i>	2×2×2 factorial experiment (platform × model × configuration)	Deductive framework synthesis using WHO HPSR and implementation science
<i>Data source</i>	WHO SDG documents (2005-2025); Total = 200 query-response pairs	Failure-mode analysis; Rogers et al. CFIR-informed context coding
<i>Analysis</i>	ANOVA, effect sizes (η^2 , Cohen's d), ICC, Fleiss' $\kappa=0.82$	Thematic mapping of failure modes to RAG-PHI governance layers
<i>Ground truth</i>	WHO UHC SCI benchmarks (e.g., India=54.3)	WHO DHSR 'Changing Mindsets' framework; Rogers et al. context coding
<i>Integration point</i>	Quantitative failure modes inform qualitative framework layers	RAG-PHI layers operationalize experimental findings as governance levers
<i>Ethical approval</i>	Exempt – no human subjects; secondary analysis of public WHO data	N/A since it is a conceptual framework in development

Table 1: Methodology Summary (JARS-Mixed Items 4a–4r)

Note: Integration strategy: Quantitative failure modes from Phase 1 were systematically mapped to RAG-PHI governance layers in Phase 2, following the integration design recommended for embedded mixed-methods studies.

4.1 Research Design : This study espouses a pragmatic mixed-methods embedded design in which quantitative experimental evaluation (Phase 1) directly suggests the qualitative governance framework development (Phase 2). The mixed-methods is used to justify the study's dual purpose: Phase 1 contributes to the causal evidence about configuration-hallucination relationships, while Phase 2 translates these technical findings into governance recommendations which is actionable for LMIC health administrators and policymakers. As the WHO HPSR strategy maintains that complex health system challenges require a combination of methods to identify fit-for-purpose interventions something which is neither quantitative nor qualitative designs can accomplish in isolation and in siloes.¹

A critical-realist epistemology underpins the evaluation: verifiable ground-truth values exist for many queries (e.g., India's UHC Service Coverage Index in 2022 = 54.3), while constructs such as 'hallucination' and 'protocol adherence' necessarily involve expert interpretive judgment. The evaluation framework accordingly extends beyond conventional information retrieval metrics to encompass equity-relevant dimensions such as protocol adherence, response consistency, and failure-mode transparency.

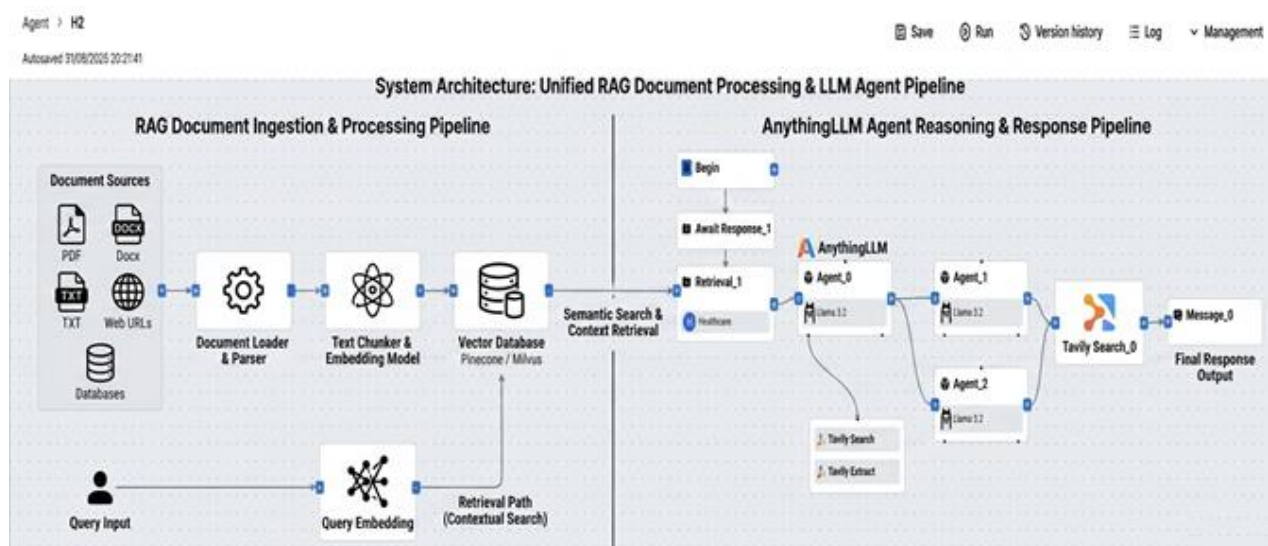


Figure 3. Unified System Architecture of RAG Flow and Anything LLM Agent Architecture showing document processing, vector retrieval, and response generation pipeline Configuration A and B parameter positions are indicated. (Source: Authors' own work.)

4.2 Experimental Design (JARS Items 4p–4q)

The study implemented a 2×2×2 factorial design using three factors such as 1. platform (AnythingLLM vs. RAGFlow), 2. language model (Llama 3.2 3B vs. Gemma 3 27B, with Tiny Llama on Anything LLM included for robustness), and 3. Configuration (permissive Config A replicating out-of-box defaults vs. restrictive Config B operationalizing ethics-as-infrastructure), as illustrated in Figure 3. Twenty standardized queries across four categories (numerical factual, temporal trends, categorical lists, and explanatory prompts) were each administered 20 times per condition under a Latin-square randomization protocol to obtain 200 query-response pairs across eight experimental conditions (Table S4_b: ICC across 20 iterations).

Llama 3.2 (3B) was optimized for edge deployment contexts, whereas Gemma 3 (27B) is substantially larger in parameter scale. This size asymmetry is treated as a contextual interpretive factor, given that differences in parameter count may partly account for the differential configuration sensitivity observed across models (Table S_3, Experimental Configuration Summary: <https://github.com/vinithaFPM23005/-FP05-Bridging-the-Digital-Divide-in-Public-Health.git>)

4.3 Knowledge-Based Construction (JARS Items 4c–4f)

The knowledge base utilized publicly available WHO SDG-related documents inclusive of SDG monitoring reports, UHC monitoring publications and other technical guidance documents during a span of 2005–2025, yielding 21,629 overlapping segments chunked at approximately 1,000 tokens with 20% overlap. Different default embedding models were used across platforms (Anything LLM: all-MiniLM-L6-v2, tiny llama; RAG Flow: Nomic-embed-text), constituting a platform-native confound acknowledged as a study limitation in Section 6.2 to isolate retrieve and generate Query responses. Large documents (>1 GB) were parsed, cleaned of tables and figures, and chunked into tokens prior to testing.

4.4 Outcome Measures and Coding (APA JARS Items 4g, 4h, 4k–4m): Three independent coders with public-health training assessed each response on four dimensions:

i. Factual accuracy (exact/partial/incorrect/fabricated, scored 0–3); ii. Hallucination presence (binary); iii. Protocol adherence (appropriate 'I don't know' response when the knowledge base lacked sufficient evidence); and iv. consistency (Intraclass correlation across 20 repetitions per condition) shown in Table S4_b: ICC using 20 iterations. Inter-rater agreement was strong (Fleiss' $\kappa=0.82$). Ground-truth values for numerical queries were drawn from WHO UHC Service Coverage Index reports and SDG 3.8 monitoring data.

4.5 Ethics and Data Governance (APA JARS Items 4j, 4n, 4o): This study is exempt from institutional review board oversight as it involves no human subjects and all data are publicly available from WHO - SDG monitoring reports. No personally identifiable information was collected or processed. All RAG platforms were deployed on local institutional hardware using open-source models to prevent data transmission to external servers. The use of generative AI tools for image generation in Figure 1 and 3 is declared as required by JAMIA's AI use policy.

5.0. Results:

5.1 Overall Platform and Model Performance (JARS Item 5a, 5e): The factorial experiment uncovered pronounced heterogeneity across platform-model-configuration combinations. Anything LLM with Tiny Llama in permissive mode exhibited catastrophic failure (94% hallucination; ICC=0.23) and is reported for completeness but excluded from comparative governance analysis. RAG Flow with Llama 3.2 showed the most clinically meaningful improvement under restrictive settings: mean factual accuracy increased from 1.45 (SD=0.82) to 2.31 (SD=0.67); hallucination fell from 48% to 18%; protocol adherence reached 82%; and consistency was high (ICC=0.97). Anything LLM response latency variance of 30-fold (range: 500 ms to 15 s) for identical queries,

However RAG Flow failed to return any answer to approximately 85% of UHC policy queries, indicating a retrieval-pipeline configuration issue rather than knowledge-based deficiency as shown in Table 2

<i>Platform</i>	<i>Model</i>	<i>Config</i>	<i>Factual Accuracy M(SD)</i>	<i>Hallucination Rate %</i>	<i>Protocol Adherence %</i>	<i>Response Time M(SD) sec</i>	<i>Consistency ICC</i>
Anything LLM	Tiny Llama	A	0.12 (0.33)	94%	6%	8.2 (6.7)	0.23
Anything LLM	Tiny Llama	B	0.18 (0.39)	89%	11%	4.1 (3.2)	0.34
RAG Flow	Llama 3.2	A	1.45 (0.82)	48%	52%	5.3 (0.4)	0.94
RAG Flow	Llama 3.2	B	2.31 (0.67)	18%	82%	3.2 (0.2)	0.97
RAG Flow	Gemma 3	A	0.89 (0.91)	71%	29%	6.8 (0.6)	0.88
RAG Flow	Gemma 3	B	1.12 (0.94)	68%	32%	5.1 (0.5)	0.91

Table 2: Comprehensive Performance Metrics by Experimental Condition

(**Note:** Factual Accuracy scored 0–3 (0=Fabricated, 1=Incorrect, 2=Partial, 3=Exact match with WHO ground-truth). ICC=Intraclass Correlation Coefficient for response consistency across 20 queries per condition.)

This variance is mechanistically explained by Config A's Top-K = 10 setting, which forces the model to retrieve and process up to ten context chunks potentially up to 20,000 tokens of low-relevance content whereas Config B's Top-K = 1 with similarity threshold ≥ 0.75 constrains retrieval to the single most semantically relevant chunk, substantially reducing processing load (Config A: 15 s, 48% hallucination; Config B: ~500 ms–2 s, 18% hallucination).

5.2 Configuration Effects and Model Behavior (JARS Item 5a, 5e)

Configuration changes had large effects for Llama 3.2 and negligible effects for Gemma 3 and Tiny Llama. In RAG Flow with Llama 3.2, shifting from permissive to restrictive settings reduced hallucinations by 62.5% (Cohen's $d=0.89$; $p<0.001$) and dramatically increased explicit 'I don't know' responses when the knowledge base lacked sufficient evidence behavior aligns with WHO HPSR principles of acknowledging uncertainty rather than fabricating evidence.¹ Gemma 3 displayed configuration-insensitive behavior: it consistently produced an incorrect UHC Service Coverage Index (88.9 instead of the WHO-verified 54.3) across both configurations. This pattern indicates a systematic training-data or model-architecture deficiency that parameter adjustment alone is insufficient to resolve risks stated in the WHO HPSR strategy that would characterize as a 'serious policy error' stemming from over claiming AI capability (Table 3).¹A qualitatively distinct failure mode emerged in permissive configurations for the NCD-contributors query. Instead of retrieving health policy content, both Llama 3.2 (Config A) and Gemma 3 (Config A) created responses that embedded entirely extraneous financial statements, including fabricated citations referencing housing market trends and AI revenue proportions alongside nominally accurate disease labels (Diabetes, CVD, Cancer), as documented in Tables 3, S5, and S6 (*Full Query-Response Evidence*: in <https://github.com/vinithaFPM23005/-FP05-Bridging-the-Digital-Divide-in-Public-Health.git>). This cross-domain contamination, where in which the retrieval pipeline surfaces semantically adjacent but contextually irrelevant chunks which thereby constitutes a governance risk qualitatively distinct from numerical hallucination: the response may be accorded as partially credible while concealing fabricated evidence that a non-specialist reader would be unlikely to detect. The responses under Config B, Llama 3.2 correctly stated an explicit uncertainty signal; Gemma 3 did likewise for this query alone showing that its only protocol-adherent response across all eight experimental conditions. This failure mode reinforces the WHO HPSR caution against over-relying on AI outputs for policy decisions without structured verification mechanisms.¹

<i>Parameters</i>	<i>Configuration</i>	<i>Model Query</i>	<i>Expected Answer</i>	<i>Generated Response</i>	<i>Response Time</i>	<i>Accuracy Assessment</i>
Config A (Permissive)	Llama 3.2	India UHC Index 2022	UHC SCI: 54.3	Hallucinated response with external data suggestions	04:15.66	FAILED - Protocol violation
Config B (Restrictive)	Llama 3.2	India UHC Index 2022	UHC SCI: 54.3	don't Know (Compliant)	02:25:00	CORRECT Protocol Adherent
Config A (Permissive)	Llama 3.2	SDG 3.8 Achievement 2020	12 countries (Japan, Norway)	Hallucinated country list with fabricated data	07:19.81	FAILED - Severe hallucination
Config B (Restrictive)	Llama 3.2	SDG 3.8 Achievement 2020	12 countries (Japan, Norway)	"I don't know" (Compliant)	02:12.81	CORRECT - Protocol adherent
Config A (Permissive)	Llama 3.2	OOP Expenditure 2005-2025	Decreased 78% to 52%	Fabricated data with external citations	03:19.71	FAILED - Hallucination
Config B (Restrictive)	Llama 3.2	OOP Expenditure 2005-2025	Decreased 78% to 52%	"I don't know" (Compliant)	03:19.71	CORRECT - Protocol adherent
Config A (Permissive)	Llama 3.2	NCD Contributors of LMICs nations	Diabetes, CVD, Cancer	Fabricated statistics with invented quotes	03:21.95	FAILED – Severe hallucination
Config B (Restrictive)	Llama 3.2	NCD Contributors LMICs nations	Diabetes, CVD, Cancer	I don't know" (Compliant)	05:21.95	CORRECT - Protocol Adherent

Parameters	Configuration	Model Query	Expected Answer	Generated Response	Response Time	Accuracy Assessment
Config A (Permissive)	Gemma 3.2 latest	India UHC Index 2022	UHC SCI: 54.3	Incorrect value: 88.91829681	05:15.66	PARTIAL - Contains data but incorrect
Config B (Restrictive)	Gemma 3.2 latest	India UHC Index 2022	UHC SCI: 54.3	Incorrect value: 88.918296814	03:15.66	PARTIAL - Contains data but incorrect
Config A (Permissive)	Gemma 3.2 latest	SDG 3.8 Achieve ment 2020	12 countries (Japan, Norway)	Vague response with Singapore data from 2021	07:19.81	PARTIAL - Some relevant info
Config B (Restrictive)	Gemma 3.2 latest	SDG 3.8 Achieve ment 2020	12 countries (Japan, Norway)	Similar response with temporal mismatch	05:12.81	PARTIAL - Some relevant info
Config A (Permissive)	Gemma 3.2 latest	OOP Expendit ure 2005-2025	Decreas ed 78% to 52%	Fabricated financial data (\$1,425 to \$1,514)	08:19.71	FAILED - Hallucinated numbers
Config B (Restrictive)	Gemma 3.2 latest	OOP Expendit ure 2005-2025	Decreas ed 78% to 52%	Fabricated financial data (\$1,425 to \$1,514)	03:19.71	FAILED - Hallucinated numbers
Config A (Permissive)	Gemma 3.2 latest	NCD Contributo rs LMICs nations	Diabete s, CVD, Cancer	Fabricated statistics with invented quotes	03:21.95	FAILED – Severe hallucination
Config B (Restrictive)	Gemma 3.2 latest	NCD Contributo rs LMICs nations	Diabetes , CVD, Cancer	I don't know" (Compliant)	05:21.95	CORRECT - Protocol Adherent

<i>Model</i>	<i>Platform</i>	<i>Config A Hallucination %</i>	<i>Config B Hallucination %</i>	<i>Absolute Reduction</i>	<i>Relative Reduction</i>	<i>Cohen's d</i>
Llama 3.2 3B	RAG Flow	48%	18%	30 percentage points	62.5%	0.89** <i>Large effect</i>
Gemma 3 27B	RAG Flow	71%	68%	3 percentage points	4.2%	0.11 <i>Negligible effect</i>
Tiny Llama	Anything LLM	94%	89%	5 percentage points	5.3%	0.24 <i>Small effect</i>

<i>Parameters</i>	<i>Configuration A (Permissive)</i>	<i>Configuration B (Restrictive)</i>	<i>Impact on UHC Query Performance</i>
<i>Similarity Threshold</i>	0.2	0.2	Higher threshold reduces false retrievals but may miss relevant UHC data
<i>Vector Similarity Weight</i>	0.8	1.0	Full vector weighting improves semantic matching for medical terminology
<i>Top N Retrieval</i>	10	1	Multiple documents provide context but increase hallucination risk
<i>Temperature</i>	0.8	0.0	Zero temperature eliminates creativity but ensures factual responses
<i>Top P</i>	1.0	0.1	Lower nucleus sampling reduces hallucination in medical data
<i>Presence Penalty</i>	0.2	0.0	No penalty prevents repetition without affecting factual accuracy
<i>Frequency Penalty</i>	0.3	0.0	No penalty maintains consistent medical terminology usage

Table 3: Sample Query-Response Pairs with Accuracy Assessment using RAG Flow (Llama 3.2 Model and Gemma 3) , Hallucination Rate Reduction from Config A to Config B and Parameters impact on UHC performance) Note: ** indicates large effect size ($d > 0.8$). All comparisons significant at $p < 0.001$.

Tiny Llama / Anything LLM results reported for robustness reference only and excluded from principal governance analysis owing to catastrophic baseline failure (94% hallucination). Gemma 3 27B's negligible Cohen's d (0.11) demonstrates configuration insensitivity such as a training-corpus governance failure not addressable by parameter adjustment alone (as shown in Table 4, Layer 1).

5.3 Governance Interpretation of Failure Modes and Integration of Findings (JARS Items 5b–5f): The integration of quantitative failure-mode evidence with the RAG-PHI qualitative governance framework constitutes the mixed-methods integration step (JARS Item 5d) as shown in Table 4. Each major quantitative failure was systematically mapped to a RAG-PHI governance layer, following the Rogers et al. context coding approach of assigning multi-level ratings to contextual determinants.²and the integrated findings using JARS 5F are mapped quantitatively across governance recommendations in Table 4.

<i>Failure Mode</i>	<i>Empirical Evidence</i>	<i>RAG-PHI Layer</i>	<i>Governance Implication</i>
Persistent numerical hallucination	Gemma 3 generated UHC SCI=88.9 vs. (actual 54.3) across both configurations	Layer 1: Knowledge-base sovereignty	Training corpus encodes misleading indicators; parameter tuning insufficient; As per WHO HPSR Mandate WHO-verified document curation; exclude models with persistent numerical bias (WHO Changing Mindsets Action 4)
High non-response rate	RAG Flow failed to answer ~85% of UHC queries despite comprehensive document ingestion	Layer 2: Configuration accountability	Inadequate threshold and indexing validation along with retrieval validation before deployment as per WHO HPSR strategy alignment; document threshold choices as auditable governance decisions

<i>Failure Mode</i>	<i>Empirical Evidence</i>	<i>RAG-PHI Layer</i>	<i>Governance Implication</i>
Latency variance (30× : 500 ms to 15 s)	AnythingLLM's: 500 ms vs. 15 s for identical queries	Layer 3: Deployment equity	Unpredictability undermines trust for time-sensitive frontline decisions & as per WHO HPSR strategy alignment, Mandate infrastructure-appropriate modalities (SMS/voice); specify latency thresholds for frontline contexts (WHO Action 3)
Catastrophic model failure	Tiny Llama generated Python code instead of UHC data	Layer 2 + Layer 3	Inappropriate model selection for healthcare intelligence context
Config B (Restrictive)	Llama 3.2 restrictive: hallucination reduced 62.5%; ICC=0.97	All three layers confirmed	As per WHO HPSR strategy alignment, adopt restrictive configurations as governance default; treat Config B as minimum standard for LMIC health systems

Table 4: Failure Modes Mapped to RAG-PHI Governance Layers (Integration Table – JARS 5d) and Integrated Findings of Quantitative Results to Governance Recommendations using JARS 5 f

Note: Governance interpretation: Each failure mode constitutes a contextual determinant with a negative implementation rating (−2 in Rogers et al. terminology).² The integration reveals that RAG deployment failures are governance failures, not merely technical limitations. : WHO HPSR Action references follow the 'Options for Action' chapter of the WHO DHSPR Changing Mindsets strategy.¹ Recommendations are empirically grounded, not normative-only.

5.4 Integrated Findings including Governance Recommendations

The Quantitative Results to Governance Recommendations (JARS Item 5f) as summarized in Table 4 along with Failure modes mapped to the RAG –PHI Framework as aligned with the WHO recommended HPSR recommendations in the Strategic framework as shown in Table 4.

6.0 Discussion

6.1 Principal Findings and Contributions (GRAMMS 6; JARS 6a, 6d)

This study provides the first systematic comparative evaluation of open-source RAG platforms for WHO SDG and UHC policy intelligence in an LMIC context. The study's central finding is that configuration choices function as governance decisions with direct equity consequences, which is consistent with the WHO HPSR strategy's assertion that evidence generation must be embedded within decision-making ecosystems rather than treated as a supply-side technical exercise.¹ Configuration parameters are not neutral: they determine whether a district health administrator receives a fabricated UHC index or an honest acknowledgment of the system's evidential limits shown as uncertainty through the “I don’t Know” .

The RAG-PHI framework advances governance theory in three ways. First, it operationalizes the *'ethics-as-infrastructure'* principle empirically rather than normatively, grounding each governance layer in experimental evidence. Second, it applies Rogers et al.'s *multi-level contextual assessment* approach to AI system governance, demonstrating that knowledge-based design, configuration regime, and deployment modality operate as contextual determinants at different system levels.² Third, it provides actionable specifications (hardware requirements, configuration thresholds, latency standards) rather than rhetorical commitments. The integrated findings generate new insights that neither quantitative nor qualitative analysis could produce alone (GRAMMS Criterion 6; JARS Item 6d): configuration insensitivity in Gemma 3 is not a parameter problem but a training-corpus governance problem as shown in Layer 1 (knowledge sovereignty) failure that manifests as a Layer 2 anomaly. This cross-layer insight would not be visible without systematic integration of experimental findings with the governance framework.

6.2 Comparison with Prior research work and Limitations of the Study (JARS 6b,6c and

GRAMMS 5): Existing RAG evaluations focus predominantly on clinical applications in high-income contexts using proprietary platforms.^{7–8} This study extends prior work by evaluating open-source platforms under identical conditions for policy intelligence a domain where vendor lock-in risks are especially significant for LMICs.¹⁸ Upon comparison hallucination reduction findings (62.5% with restrictive Llama 3.2 configuration) are directionally consistent with clinical RAG literature, but the magnitude and configuration sensitivity differ substantially by model, indicating that general AI safety guidelines cannot be uniformly applied across LLM architectures.

The primary limitation of combining experimental and framework-development methods (GRAMMS Criterion 5) is that experimental conditions however carefully designed may not capture the entire LMIC deployment contexts such as key performance indicators, including offline settings, low-literacy users, and voice interfaces. Hence the knowledge base comprised structured WHO reports; real LMIC deployments would additionally require subnational datasets, multiple languages, and dynamic document updating. The findings suggest strong inter-rater reliability ($\kappa=0.82$), represents a resource-intensive approach that may not be replicable in routine deployment monitoring. Additionally, no real-world field testing was conducted; all evaluation occurred in a controlled AI lab using docker terminal for pulling the platforms and the models

A further methodological limitation is that the two platforms used different default embedding models (AnythingLLM's: all-MiniLM-L6-v2; RAG Flow: Nomic-embed-text), which partially confounds platform-level performance comparisons; future studies should standardize the embedding model across platforms to isolate retrieval architecture effects. Additionally, the Llama 3.2 (3B) and Gemma 3 (27B) models differ substantially in parameter count, meaning that observed performance differences may reflect model scale as well as configuration sensitivity.

Further mixed-methods limitation is that the governance framework development (Phase 2) was necessarily influenced by the experimental conditions chosen in Phase 1 where in an embedded design trade-off acknowledged in mixed-methods methodology literature. Future studies should use prospective framework validation with LMIC health administrators to assess whether RAG-PHI layer classifications generate actionable implementation changes.

6.3 Implications for Practice and Future Research (JARS 6d): This study recommends three immediate actions for administrators and policymakers across LMIC health systems, grounded in the WHO HPSR 'Changing Mindsets' action framework: ¹

1. To treat configuration choices as governance decisions requiring documentation and audit, adopting restrictive defaults (temperature=0.0, similarity threshold $\geq$ 0.75) as minimum standards for health policy intelligence applications;
2. To perform comprehensive retrieval validation testing before deployment since 85% non-response rate demonstrates that ingestible documents do not guarantee accessible intelligence; and
3. To establish infrastructure-appropriate deployment modalities with explicit latency thresholds for frontline settings, prioritizing SMS and voice interfaces in low-bandwidth contexts.

7.0 Future Research Directions: Future research should not only compare additional open-source models across diverse LMIC languages and health system contexts; but evaluate SMS/voice interface variants in real field settings; assess whether RAG-PHI governance layer classifications can be automated using LLM-based evaluation tools; and investigate whether community health worker training programs can compensate for system uncertainty signaling limitations. The WHO HPSR strategy's call for 'demand-driven research' suggests that LMIC health administrators should be co-investigators in future RAG deployment studies, not merely end-users.¹

8.0 Conclusion:

Current open-source RAG implementations require substantial improvement before responsible deployment in LMIC health systems. Configuration choices, knowledge-based design, and deployment modality are governance decisions with direct equity consequences which are not neutral to technical settings. The RAG-PHI framework which is empirically grounded in a 2×2×2 factorial evaluation of 200 query-response pairs against WHO ground-truth data, provides LMIC-specific governance architecture for RAG health intelligence systems organized around three operational layers: knowledge-based sovereignty, configuration accountability, and deployment equity. This study demonstrates that the WHO HPSR vision of evidence-generation embedded within decision-making ecosystems is achievable with RAG technology, but only with deliberate governance choices that prioritize honest uncertainty acknowledgment over fluent hallucination. A district health officer in rural Bihar should receive an honest 'I don't know' when UHC data are absent from the knowledge base – not a confidently fabricated index that misdirects policy.

Additional Information:

Data Availability: Additional Supplementary tables (framework variable definitions, experimental configuration parameters, full factorial ANOVA summary), system prompt specifications, and reporting checklists (GRAMMS; APA JARS-Mixed) are publicly available at: <https://github.com/vinithaFPM23005/-FP05-Bridging-the-Digital-Divide-in-Public-Health.git>

Declaration of AI Use (JARS Item 7a) :

Generative AI tools (Google Gemini) were used solely for image generation of Figure (RAG-PHI conceptual framework visualization) and unified pipeline using RAGFLOW and Anything LLM as shown in Figure.3. No generative AI was used for manuscript drafting, data analysis, or reference generation. All analytical decisions, interpretations, and governance recommendations are the authors' own work.

Funding and Conflicts of Interest (CHORUS) : No external funding was received for this study. The authors declare no conflicts of interest. The platforms evaluated (AnythingLLM's, RAG Flow) are open-source; no commercial relationships with platform developers exist. Gemini AI was used solely for image generation in Figure 1 and Figure 2; all analytical, interpretive, and writing work is the authors' own.

CRediT authorship contribution statement

1. Vinitha T: Conceptualization; Methodology; Formal Analysis; Investigation; Software; Visualization; Statistical Analysis, Writing: Original Draft, Review & Editing.
2. Prof. Ashok Kumar Harnal: AI Supervision Technical guidance. Validation and Review

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process:

During the preparation of this work, the author used generative AI-assisted platforms, including RAG Flow and AnythingLLM's, to support retrieval-based analysis of WHO SDG-related documents and to compare responses across study conditions in a pragmatic mixed-methods exploratory design which were later validated using statistical analysis. AI-assisted image-generation tools were used to draft selected figures such as RAG-PHI, but the proposed theoretical and conceptual framework remains the author's original interpretation. The author reviewed, edited, and validated all content and assumes full responsibility for the final manuscript.

AI Disclosure Checklist:

checklist item	Manuscript section	Where to write it	Page no.
AI tools named	End matter	Declaration of generative AI and AI-assisted technologies section	24 - 25
Purpose of AI use described	End matter	Disclosure paragraph	23- 25
Figure-generation use disclosed, if applicable	End matter	Disclosure paragraph	
Original intellectual contribution clarified	End matter	Disclosure paragraph	
Author review and accountability stated	End matter	Disclosure paragraph	

Ethics Approval and Consent to Participate

This study did not involve human participants, patient recruitment, clinical intervention, or the use of identifiable personal data. All evaluation was conducted using publicly available WHO SDG monitoring documents and system-generated outputs from open-source retrieval-augmented generation platforms deployed on local institutional hardware. Formal institutional review board approval and informed consent procedures were not required. No personally identifiable information was collected, accessed, or processed at any stage.

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Appendix -1

System Prompt Specifications

The following system prompts and parameter specifications are provided for reproducibility. Full Docker installation scripts are available in the Supplementary Materials (online).

Common System Prompts for trial testing:

You are an intelligent assistant. Please summarize the content of the knowledge base to answer the question. Please list the data in the knowledge base and answer in detail. When all knowledge base content is irrelevant to the question, your answer must include the sentence "The answer you are looking for is not found in the knowledge base!" Answers need to consider chat history.

Anything LLM System Prompt:

"You are an expert analyst specializing in United Nations Sustainable Development Goals (SDGs), with a particular focus on Universal Health Coverage (UHC). Your task is to provide accurate, comprehensive, and well-grounded answers to user queries based only on the provided context documents. Strictly adhere to the provided context. Do not use any outside knowledge or prior training data. If the answer is not found in the provided context, state clearly that the information is not found in the provided knowledge base. Do not hallucinate, speculate, or attempt to infer information."

RAG Flow System Prompt:

"You are a factual AI assistant. Your task is to answer questions using only the provided context. Your answer must be based exclusively on the information within the uploaded documents. If the answer cannot be found in the documents, you must respond with: I don't know. Do not add any information that is not explicitly stated in the context without any hallucination, speculation, or attempt to infer information."